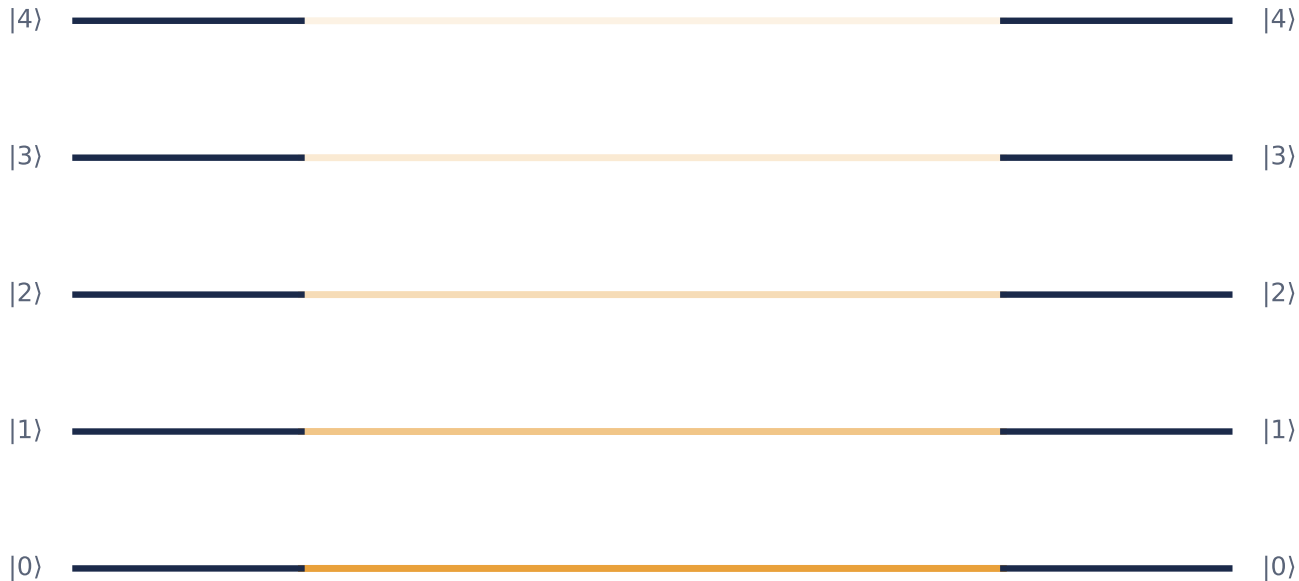


L

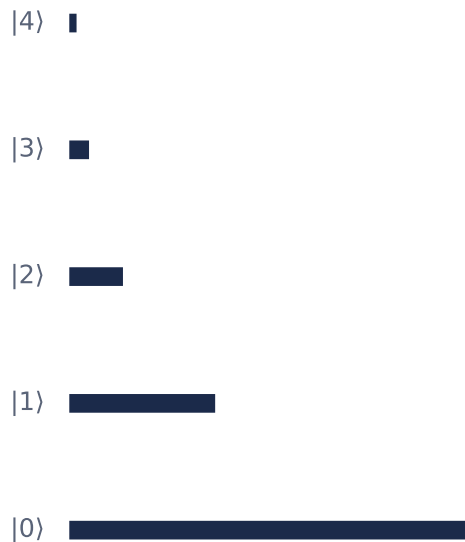
R



link opacity $\propto e^{(-\beta E_n/2)}$

$$|\text{TFD}(\beta)\rangle = Z^{-1/2} \sum_n e^{(-\beta E_n/2)} |n\rangle_L |n\rangle_R$$

Tr_R



$$\rho_L = e^{(-\beta H)}/Z$$

a thermal state at temperature $1/\beta$